long, crest to crest, to high-frequency gamma rays no longer than a quadrillionth of a meter. That rich palette of light supplies no end of astrophysical discoveries: Curious how much gas lurks among the stars in galaxies? Radio telescopes do that best. There is no knowledge of the cosmic background, and no real understanding of the big bang, without microwave telescopes. Want to peek at stellar nurseries deep inside galactic gas clouds? Pay attention to what infrared telescopes do. How about emissions from the vicinity of ordinary black holes and supermassive black holes in the center of a galaxy? Ultraviolet and X-ray telescopes do that best. Want to watch the high-energy explosion of a giant star, whose mass is as great as forty suns? Catch the drama via gamma ray telescopes.

We've come a long way since Herschel's experiments with rays that were "unfit for vision," empowering us to explore the universe for what it is, rather than for what it seems to be. Herschel would be proud. We achieved true cosmic vision only after seeing the unseeable: a dazzlingly rich collection of objects and phenomena across space and across time that we may now dream of in our philosophy.

† Not until the mid-1800s, when the physicist's spectrometer was applied to astronomical problems, did the astronomer become the astrophysicist. In 1895, the prestigious *Astrophysical Journal* was founded, with the subtitle "An International Review of Spectroscopy and Astronomical Physics."

†† William Herschel, "Experiments on Solar and on the Terrestrial Rays that Occasion Heat," *Philosophical Transactions of the Royal Astronomical Society*, 1800, 17.

††† Antonie van Leeuwenhoek, letter to the Royal Society of London, October 10, 1676.

†††† All waves follow the simple equation: *speed = frequency* × *wavelength*. At a constant speed, if you increase the wavelength, the wave itself will have smaller frequency, and vice versa, so that when you multiply the two quantities you recover the same speed of the wave every time. Works for light, sound, and even fans doing the "Wave" at sports arenas—anything that's a traveling wave.

††††† Karl Jansky, "Electrical Disturbances Apparently of Extraterrestrial Origin," *Proceedings of the Institute for Radio Engineers* 21, no. 10 (1933): 1387.

†††††† "pico-" is the metric prefix for one-trillionth.